

Keshav Sharma

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LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**
B.Tech in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++
Data Analytics & Visualization: Power BI, DAX, Pandas, NumPy, Matplotlib, Plotly, Exploratory Data Analysis, Statistical Analysis
Financial & Risk Analytics: Financial Data Analysis, Transaction Analytics, Risk Analysis, Anomaly Detection, Time-Series Analysis, Statistical Outlier Analysis, Customer Segmentation
Machine Learning: Scikit-learn, Isolation Forest, LSTM Autoencoders, Feature Engineering, Model Evaluation, Supervised & Unsupervised Learning
Data Engineering & Databases: SQLite, MySQL, MongoDB, SQL Data Modelling, ETL/Data Pipelines, Data Validation, Data Quality
AI & Automation: Generative AI, LLMs, Google Gemini API, NLP, OCR, Process Automation, REST APIs
Tools & Engineering: Git/GitHub, FastAPI, Flask, Streamlit, PyTest, Jupyter Notebook, Reproducible Experimentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic-search concepts involving document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM-generated responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%) using context-aware hourly consumption patterns.
- Implemented automated email alerts and responsible AI controls, including transparent anomaly scoring, local data processing, privacy safeguards, and rate-limited notifications capped at 1 alert/hour.

VOIS – Vodafone Idea Foundation virtual internship* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end financial analytics pipeline processing 50K+ PaySim transactions, combining statistical outlier analysis, duplicate detection, vendor-velocity checks, HHI concentration analysis, and Isolation Forest for transaction-risk detection.
- Designed a star-schema SQLite data warehouse and composite 0–100 risk scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures to surface vendor risk, anomaly rates, transaction trends, and risk insights, supported by automated pipeline tests.

- Implemented data validation and control checks across transaction, vendor, and risk datasets, improving auditability and ensuring consistent outputs across the analytics pipeline.

Real-Time Stock Market Anomaly Detector **|LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.
- Integrated time-series feature engineering and interactive Plotly/Streamlit visualisations to monitor anomaly patterns, market signals, and model outputs for analytical review.

Customer Lifetime Value Engine **|LINK**

- Built a probabilistic model using BG/NBD and Gamma-Gamma on 1M+ historical transaction records to forecast customer lifetime value and purchasing behaviour.
- Segmented customers into value-based cohorts to identify which segments drive the most revenue, supporting prioritisation.
- Identified the top 20% of customers driving ~65% of forecasted revenue, visualised through Power BI dashboards for stakeholder-facing reporting.
- Performed transaction-level preprocessing, frequency/recency analysis, and model validation to improve the reliability of customer-level value estimates and segmentation.

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.
- Automated structured expense extraction and categorisation from receipt data, enabling faster financial record processing and reducing manual data-entry effort.

Energy Anomaly Detector (AI for Sustainability internship final project) **| LINK**

- Engineered an end-to-end Python pipeline using Isolation Forest to detect abnormal electricity consumption from hourly, time-of-day and day-of-week usage patterns.
- Developed context-aware feature engineering with hourly aggregation, rolling-average baselines, and temporal features to distinguish unusual consumption from normal usage patterns.
- Implemented automated email alerts and visual anomaly reports, flagging 9 anomalies across 6,918 test hours while limiting notifications to a maximum of one alert per hour.
- Applied responsible AI practices through transparent anomaly scoring, local data processing, privacy safeguards, and alert-rate limiting to reduce notification fatigue.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025
- Accenture Software Engineering Job Simulation | **LINK** 2025
- Prepare Data for ML APIs on Google Cloud Skill Badge |**LINK** 2026

Extracurricular Activities

Guest Committee Member | TechnoVIT, VIT Chennai

- Served as a Guest Committee Member for TechnoVIT, supporting guest communication, scheduling, and on-ground arrangements for VIT Chennai's flagship technology event.